

Integrable Systems, Transfer Matrices, and the Bethe Ansatz

Zhen Wang [based on 1012.3982]

CUHK (Shenzhen), SSE

September 11, 2026

Classical and Quantum Integrability

Transfer Matrix Formulation

Bethe Ansatz

Classical integrability

- Harmonic oscillator

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2$$

- Canonical equations

$$\dot{x} = \frac{\partial H}{\partial p} = \frac{p}{m}, \quad \dot{p} = \frac{\partial H}{\partial x} = -m\omega^2 x$$

$$\Rightarrow x(t) = A \cos(\omega t + \varphi)$$

- In the phase space, the energy is conserved

$$H(x, p) = E$$

Classical integrability

- Poisson bracket

$$\{A, B\} = \sum_{i=1}^N \left(\frac{\partial A}{\partial q_i} \frac{\partial B}{\partial p_i} - \frac{\partial A}{\partial p_i} \frac{\partial B}{\partial q_i} \right)$$

- The phase space for a general Hamilton system $H(q_i, p_i)$ has dimension $2N$
- Energy conservation

$$\frac{dH}{dt} = 0$$

reduces the phase space dimension to $2N - 1$

- For $N > 1$, extra conserved quantities commute with each other are needed
- Integrability: family of N conserved quantities that Poisson commute with each other

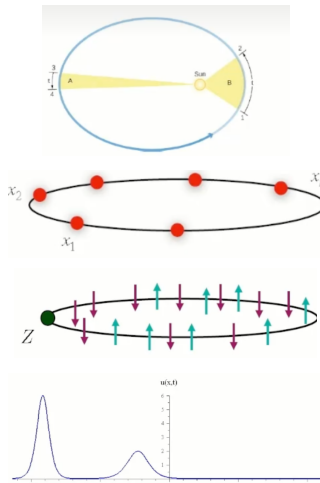
$$I_i = c_i, \quad \{I_i, I_j\} = 0, \quad i, j = 1, \dots, N$$

- Liouville–Arnold theorem: the remaining N -dimensional manifold $M \cong T^N$

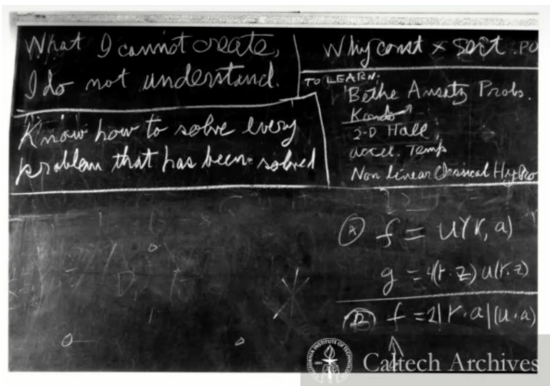
Examples

- Easy problems from school/undergrad
i.e. oscillator, Kepler problem, etc.
- Many-body particle systems
- Spin Chains
- Korteweg–de Vries, integrable PDE

$$u_t = 6uu_x - u_{xxx}$$



Classical integrability



I got really fascinated by these $(1+1)$ -dimensional models that are solved by Bethe ansatz and how mysteriously they jump out at you and work and you don't know why. I'm trying to understand all this better.

Quantum integrability and the spin chain

- An integrable quantum system possesses a large family of conserved charges $Q_1, Q_2, \dots$ such that $[Q_r, Q_s] = 0$
- Integrability of spin chain $\longleftrightarrow$ integrability of $\mathcal{N} = 4$ SYM (probably the topic for the next time!)

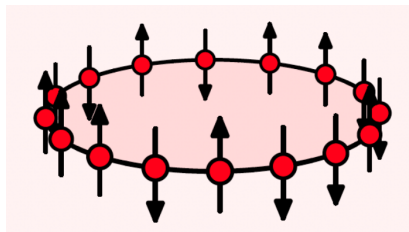
Quantum integrability and the spin chain

- Heisenberg XXX spin chain:

$$H_{XXX} = J \sum_{l=1}^L \mathbf{S}_l \cdot \mathbf{S}_{l+1} = J \sum_{l=1}^L (\mathbf{S}_l^x \mathbf{S}_{l+1}^x + \mathbf{S}_l^y \mathbf{S}_{l+1}^y + \mathbf{S}_l^z \mathbf{S}_{l+1}^z)$$

- Periodic boundary condition:

$$\mathbf{S}_{L+1} \equiv \mathbf{S}_1$$



Quantum integrability and the spin chain

- The nearest-neighbor permutation operator exchanges the two spin states:

$$P_{l,l+1}|A_l B_{l+1}\rangle = |B_l A_{l+1}\rangle, \quad P_{l,l+1}^2 = \mathbf{1}_{l,l+1}$$

- Define

$$\begin{aligned} |\uparrow\uparrow\rangle &= |\uparrow\rangle \otimes |\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, & |\uparrow\downarrow\rangle &= |\uparrow\rangle \otimes |\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \\ |\downarrow\uparrow\rangle &= |\downarrow\rangle \otimes |\uparrow\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, & |\downarrow\downarrow\rangle &= |\downarrow\rangle \otimes |\downarrow\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \end{aligned}$$

Quantum integrability and the spin chain

- Thereby

$$\boldsymbol{\sigma}_l \cdot \boldsymbol{\sigma}_{l+1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 2 & 0 \\ 0 & 2 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad P_{l,l+1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\Rightarrow \quad \mathbf{S}_l \cdot \mathbf{S}_{l+1} = \frac{1}{2} P_{l,l+1} - \frac{1}{4} \mathbf{1}_{l,l+1}$$

- The ferromagnetic convention used for the Hamiltonian is

$$H_{\text{XX}} = \sum_{l=1}^L (\mathbf{1} - P_{l,l+1})$$

Yang–Baxter Equation

- For a spin chain with L sites, the physical Hilbert space is

$$\mathcal{H} = V_1 \otimes \cdots \otimes V_L$$

Each V_j is the local quantum space. For a spin-1/2 system, $V_j \cong \mathbb{C}^2$

- Let V be the local spin space. An R -matrix is a two-site operator $R(u)$ depending on an auxiliary complex spectral parameter u
- XXX chain $\longrightarrow$ Yang–Baxter equation (YBE)

$$R_{12}(u-v)R_{13}(u-w)R_{23}(v-w) = R_{23}(v-w)R_{13}(u-w)R_{12}(u-v)$$

Transfer matrix formulation

- Introduce an auxiliary space $V_a \simeq V$ and define the monodromy and transfer matrices by

$$M_a(u) = R_{aL}(u) \cdots R_{a1}(u), \quad T(u) = \text{Tr}_a M_a(u)$$

- The auxiliary space is not an additional physical spin

Transfer matrix formulation

- With two auxiliary spaces V_a and V_b , the YBE gives

$$R_{ab}(u-v)R_{aj}(u)R_{bj}(v) = R_{bj}(v)R_{aj}(u)R_{ab}(u-v)$$

- Furthermore

$$R_{ab}(u-v)M_a(u)M_b(v) = M_b(v)M_a(u)R_{ab}(u-v)$$

- Assuming that $R_{ab}(u-v)$ is invertible, cyclicity of the traces gives

$$\begin{aligned} T(u)T(v) &= \text{Tr}_{a,b}(M_a(u)M_b(v)) \\ &= \text{Tr}_{a,b}\left(R_{ab}(u-v)^{-1}M_b(v)M_a(u)R_{ab}(u-v)\right) \\ &= \text{Tr}_{a,b}(M_b(v)M_a(u)) \\ &= T(v)T(u) \end{aligned}$$

Transfer matrix formulation

- Define

$$\log T(u) = \sum_{n=0}^{\infty} (u - u_*)^n Q_{n+1}$$

Given $[T(u), T(v)] = 0$, this gives commuting charges

$$Q_{n+1} = \frac{1}{n!} \frac{d}{du^n} \log T(u) \Big|_{u=u_*}$$

- Why $\log T(u)$? To fix the normalization $T'(u)/T(u)$

Transfer matrix formulation

- R-matrix renormalization

$$R_{aj} = u\mathbf{1}_{aj} + iP_{aj}$$

At the regular point $u = 0$,

$$R_{aj}(0) = iP_{aj}, \quad R'_{aj}(0) = \mathbf{1}_{aj}$$

$$M_a(0) = i^L P_{aL} \cdots P_{a1}, \quad T(0) = i^L U, \quad U = \text{Tr}_a(P_{aL} \cdots P_{a1})$$

- Immediately $Q_i = -\log U$

Transfer matrix formulation

- For $r \geq 1$, define the local XXX charges as

$$Q_r = -\frac{i}{(r-1)!} \frac{d^{r-1}}{du^{r-1}} \log \left[\frac{T(u) U^{-1}}{(u+i)^L} \right] \Big|_{u=0}$$

- Calculations give (see backup slides)

$$Q_2 = H_{\text{XXX}} \rightarrow \boxed{\text{abundant conserved quantities!}}$$

that determines the energy, and

$$Q_3 = -\frac{i}{2} \sum_{l=1}^L [P_{l,l+1}, P_{l+1,l+2}]$$

Bethe ansatz: one magnon

- Choose the ferromagnetic reference state

$$|\Omega\rangle = |\uparrow\uparrow\cdots\uparrow\rangle, \quad H_{\text{XX}}|\Omega\rangle = \sum_{l=1}^L (1 - P_{l,l+1}) |\Omega\rangle = 0$$

- A single flipped spin at site n defines a magnon

$$|n\rangle = |\uparrow\cdots \underbrace{\downarrow}_{n} \cdots \uparrow\rangle$$

- Translational invariance suggests a plane-wave excitation

$$|p\rangle = \sum_{n=1}^L e^{ipn} |n\rangle$$

- Acting with the Hamiltonian gives

$$H_{\text{XX}}|p\rangle = E(p)|p\rangle, \quad E(p) = 4 \sin^2 \frac{p}{2} \quad \boxed{\text{dispersion relation}}$$

Bethe ansatz: two magnons

- Consider two flipped spins at positions $n_1 < n_2$:

$$|\Psi\rangle = \sum_{n_1 < n_2} \Psi(n_1, n_2) |n_1, n_2\rangle$$

- Bethe ansatz: the magnons behave as plane waves

$$\Psi(n_1, n_2) = A_{12} e^{i(p_1 n_1 + p_2 n_2)} + A_{21} e^{i(p_2 n_1 + p_1 n_2)}$$

- Consequently, when the two magnons scatter,

$$A_{21} = S(p_1, p_2) A_{12}, \quad S(p_1, p_2) = -\frac{1 + e^{i(p_1 + p_2)} - 2e^{ip_2}}{1 + e^{i(p_1 + p_2)} - 2e^{ip_1}}$$

- The momenta are preserved; the interaction only produces a scattering factor

Periodic boundary condition and Bethe equations

- For M magnons, transport the k -th magnon once around the periodic chain. During this process it scatters through all other magnons:

$$e^{ip_k L} = \prod_{\substack{j=1 \\ j \neq k}}^M S(p_k, p_j)$$

- For the XXX spin chain, introduce the rapidity

$$u_k = \frac{1}{2} \cot \frac{p_k}{2}, \quad e^{ip_k} = \frac{u_k + \frac{i}{2}}{u_k - \frac{i}{2}}$$

$$\left(\frac{u_k + \frac{i}{2}}{u_k - \frac{i}{2}} \right)^L = \prod_{\substack{j=1 \\ j \neq k}}^M \frac{u_k - u_j + i}{u_k - u_j - i}$$

Bethe equations, $k = 1, \dots, M.$

Bethe roots and the spectrum

- The solutions of the Bethe equations

$$\{u_1, u_2, \dots, u_M\}$$

are called the Bethe roots

- Each root determines the momentum of one magnon:

$$e^{ip_k} = \frac{u_k + \frac{i}{2}}{u_k - \frac{i}{2}}$$

- The spectrum:

$$E_k = \frac{1}{u_k^2 + \frac{1}{4}}$$

Far better than brute-force diagonalization!

Summary

- Integrability is characterized by the existence of many conserved quantities, which constrain the dynamics
- In the XXX spin chain, the YBE generates an infinite tower of commuting charges through the transfer matrix
- The Bethe ansatz exploits factorized scattering to reduce the many-body spectrum problem to solving equations for Bethe roots

Thanks for your attention!

Derivations of the charge Q_2

- The XXX chain is generated by the rational R -matrix

$$R_{aj}(u) = u \mathbf{1}_{aj} + iP_{aj}$$

where P_{aj} exchanges the auxiliary and quantum spaces:

$$P_{aj}|\alpha\rangle_a|\beta\rangle_j = |\beta\rangle_a|\alpha\rangle_j$$

- The monodromy and transfer matrices are

$$M_a(u) = R_{aL}(u) \cdots R_{a1}(u),$$

$$T(u) = \text{Tr}_a M_a(u)$$

- The conserved charges are generated by the logarithmic expansion

$$Q_r = -\frac{i}{(r-1)!} \frac{d^{r-1}}{du^{r-1}} \log \left[\frac{T(u) U^{-1}}{(u+i)^L} \right]_{u=0}$$

Derivations of the charge Q_2

- At the regular point

$$R_{aj}(0) = iP_{aj}, \quad R'_{aj}(0) = \mathbf{1}_{aj}$$

Therefore

$$M_a(0) = i^L P_{aL} \cdots P_{a1}$$

Taking the auxiliary trace gives the cyclic shift

$$T(0) = i^L U$$

- Differentiating the monodromy product:

$$T'(u) = \sum_{j=1}^L \text{Tr}_a(R_{aL} \cdots R'_{aj} \cdots R_{a1})$$

At $u = 0$ only one factor is differentiated:

$$T'(0) = i^{L-1} U \sum_{l=1}^L P_{l,l+1}$$

Derivations of the charge Q_2

- The auxiliary spin $P_{aL} \cdots P_{a1}$ passes through the whole chain.
- The trace closes the auxiliary line and produces a lattice translation:

$$\text{Tr}_a(P_{aL} \cdots P_{a1}) = U$$

- When one R -matrix is differentiated, the corresponding permutation is removed:

$$R'_{aj}(0) = 1$$

The remaining operator differs from the shift by one nearest-neighbor exchange:

$$X_j = UP_{j-1,j}$$

Hence

$$T'(0) = i^{L-1} U \sum_{l=1}^L P_{l,l+1}$$

$$T(0)^{-1} T'(0) = -i \sum_{l=1}^L P_{l,l+1}$$

Derivations of the charge Q_2

- The second charge is

$$Q_2 = -i \left[T(0)^{-1} T'(0) - \frac{L}{i} \right]$$

Substituting

$$T(0)^{-1} T'(0) = -i \sum_{l=1}^L P_{l,l+1},$$

gives

$$Q_2 = -i \left(-i \sum_l P_{l,l+1} + iL \right) = \sum_{l=1}^L (1 - P_{l,l+1}) = H_{\text{XXX}}$$